# Project Design Phase-II

## Solution Requirements Document (Functional&Non-Functional)

| | | | |
|---|---|---|---|
| **Date** | 4 Aug 2026 | **Project Name** | Book A Doctor |
| **Student Name** | Navendhiran C | **Maximum Marks** | 4 Marks |

**Overview:** This document lays out the complete functional and non-functional requirements for the Book A Doctor platform. The system is designed to connect patients with healthcare providers through a single, streamlined experience covering appointment scheduling, doctor discovery, and consultation management.

## 1. Functional Requirements (FR)

The functional requirements outline the core capabilities, essential system features, and the workflows users step through while interacting with the platform.

| FR No. | Functional Requirement (Epic | )Sub Requirement (Story / Sub-Task) |
|---|---|---|
| **FR-1** | **User Registration** | • Sign-up through a custom form (Name, Email, Phone, Password)<br>• Sign-up via Gmail OAuth Single Sign-On (SSO)<br>• Sign-up via LinkedIn Developer Identity API |
| **FR-2** | **User Authentication & Confirmation** | • Account confirmation through a secure activation link emailed to the user<br>• Identity check via a 6-digit time-based mobile OTP<br>• Two-Factor Authentication (2FA) for healthcare professionals |
| **FR-3** | **Doctor Search & Discovery** | • Search by specialty, clinical condition, or doctor name<br>• Geo-location and distance-based search for the closest clinics<br>• Multi-criteria filtering by consultation fee, rating, and availability |
| **FR-4** | **Appointment Booking & Management** | • Live calendar slot selection with an immediate hold<br>• Workflows for rescheduling and cancelling appointments<br>• Automatic SMS and email reminders ahead of upcoming visits |
| **FR-5** | **Medical History & Records Upload** | • Patient portal for uploading prescriptions, lab results, and scans<br>• Access controls restricting patient files to the assigned doctor |
| **FR-6** | **Payment Gateway Integration** | • Support for UPI, credit/debit cards, and net banking<br>• Automatic digital receipt generation and refund handling |

## 2. Non-Functional Requirements (NFR)

The non-functional requirements set out quality benchmarks, performance targets, regulatory obligations, and system-level constraints.

| NFR No. | Non-Functional Require | mDeenstcription & Acceptance Criteria |
|---|---|---|
| **NFR-1** | **Usability** | A clean, mobile-responsive interface that lets patients finish a booking in 3 taps or fewer, conforming to WCAG 2.1 Level AA accessibility guidelines. |
| **NFR-2** | **Security** | AES-256 encryption for health data at rest and TLS 1.3 for data in transit, with full adherence to HIPAA and applicable local data-protection regulations. |
| **NFR-3** | **Reliability** | A dependable booking engine targeting 99.9% successful transaction completion, with database transaction locking preventing any double-booking. |
| **NFR-4** | **Performance** | Doctor-availability queries respond in under 1.5 seconds, and the system sustains 500 requests per second at peak load without degrading. |
| **NFR-5** | **Availability** | 99.95% cloud availability backed by automated multi-region failover, with maintenance strictly confined to 02:00-04:00 AM IST. |
| **NFR-6** | **Scalability** | A horizontally auto-scaling architecture that can absorb bursts of up to 25,000 concurrent users and 100,000 daily appointment searches. |
| **NFR-7** | **Maintainability** | A modular, containerized microservices setup (Docker/Kubernetes) enabling independent deployments and ongoing updates. |
| **NFR-8** | **Data Integrity** | Daily automated backups retained for 30 days, with point-in-time recovery (PITR) and a recovery point objective (RPO) under 5 minutes. |

## System Compliance & Compliance Note

Every functional and non-functional requirement above is aligned with the Phase-II evaluation criteria. Verification during implementation will rely on automated end-to-end integration tests together with independent third-party security audits.

Prepared By: Project Development Team

Approved By: Project Evaluator / Guide